

LECTURE 7

Forecasting and Model Selection: Evaluation, Robustness and Auto- matic Methods

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Agenda

- 01 Forecast evaluation metrics
- 02 Diebold–Mariano test
- 03 Forecasting in changing environments
- 04 Intercept correction and differencing
- 05 General-to-Specific (GETS) model selection
- 06 Autometrics and Impulse Indicator Saturation
- 07 LASSO and penalised methods
- 08 Forecast combination and machine learning

This lecture integrates material from Clements & Hendry (1998, 1999, 2011), Hendry & Doornik (2014), and Makridakis et al. (2018). The UK money demand example runs throughout.

Section 1

Forecast Evaluation

Metrics and the MSFE Optimality Criterion I

For h -step ahead forecast $\tilde{Y}_{T+h|T}$ evaluated against Y_{T+h} :

$$\text{MSFE}(h) = \frac{1}{H} \sum_{t=1}^H (Y_{T+t+h} - \tilde{Y}_{T+t+h|T+t})^2$$

The **conditional expectation** $\hat{Y}_{T+h|T} = \mathbb{E}[Y_{T+h}|\mathcal{I}_T]$ is the unique MSFE-minimising forecast in a stationary environment.

Metric	Formula	Note
RMSFE	$\sqrt{\text{MSFE}}$	Scale-dependent
MAE	$\frac{1}{H} \sum e_{t+h} $	Robust to outliers
MAPE	$\frac{1}{H} \sum e_{t+h} / Y_{t+h} $	Scale-free; undefined if $Y = 0$
Theil U	$\text{RMSFE}/[\text{RMSFE}_{\text{naive}}]$	< 1 : beats random walk

MSE decomposition (Theil, 1961):

$$\text{MSE} = \underbrace{(\bar{\tilde{Y}} - \bar{Y})^2}_{\text{bias}} + \underbrace{(s_{\tilde{Y}} - s_Y)^2}_{\text{variance}} + \underbrace{2(1 - \rho_{\tilde{Y}, Y})s_{\tilde{Y}}s_Y}_{\text{covariance}}$$

- A good forecast has most MSE in the **covariance** proportion (irreducible uncertainty)
- Large **bias** proportion: model has systematic prediction error — intercept correction needed
- Large **variance** proportion: forecast does not track fluctuations well

Recursive (expanding window): Re-estimate on $[1, t]$ before forecasting $t + h$. Appropriate when parameters are stable.

Rolling (fixed window): Use only the most recent W observations. Better when there are structural breaks near the forecast origin.

Section 2

Testing Forecast Accuracy

The Diebold–Mariano Test I

Compare predictive accuracy of two competing models A and B. Define the loss differential:

$$d_t = L(e_t^A) - L(e_t^B)$$

where $L(\cdot)$ is a loss function (typically squared or absolute error).

H_0 : equal predictive accuracy $\Rightarrow \mathbb{E}[d_t] = 0$.

Test statistic: $DM = \bar{d} / \sqrt{\hat{V}(\bar{d})} \overset{a}{\sim} N(0, 1)$

where $\hat{V}(\bar{d})$ uses HAC (Newey–West) standard error.

Key features

- Non-nested models: model A need not nest B
- No requirement that models be estimated
- Works for any loss function, any forecast horizon
- Harvey, Leybourne & Newbold (1997) small-sample

At long horizons ($h > 1$), the loss differentials are **serially correlated** by construction (overlapping forecast errors). Use HAC with bandwidth $\geq h$ —

The Diebold–Mariano Test II

Forecast combination as a benchmark

Before comparing models, always compare against:

- **Random walk:** $\tilde{Y}_{T+1} = Y_T$ (no-change benchmark)
- **Historical mean:** $\tilde{Y}_{T+1} = \bar{Y}$ (mean-reversion benchmark)
- **Equal-weight combination:** average of all model forecasts

Combining forecasts from diverse models typically outperforms any individual model in MSFE terms, even if the individual models are misspecified in different ways. The gains are largest when models are structurally different and have low pairwise correlation of errors.

Optimal combination weights

$\tilde{Y} = w\tilde{Y}_A + (1 - w)\tilde{Y}_B$. OLS regression of Y_{t+h} on $\tilde{Y}^A$ and $\tilde{Y}^B$ gives optimal $\hat{w}$. But equal weights often perform nearly as well (Makridakis & Winkler, 1983).

Section 3

Forecasting in Changing Environments

Why do forecasts fail? Taxonomy of forecast errors (Clements & Hendry, 1999):

- 1 **Innovation variance** σ^2 : irreducible — cannot be forecast
- 2 **Estimation error** ($O(T^{-1/2})$): shrinks with sample size; small for $T > 100$
- 3 **Incorrect model** (model misspecification): systematic bias
- 4 **Location shift at forecast origin**: a change in $\mu_y = \mu/(1 - \alpha)$ at time T produces **systematic** forecast errors — this is the **primary** cause of large forecast failures

Simulations in Clements & Hendry (1999) show that location shifts at the forecast origin are **far more damaging** than other sources of error. A change in α that leaves μ_y unchanged causes only small MSFE increases. But a change in μ_y causes immediate and growing forecast bias.

Forecast Failure and Robust Methods II

Intercept correction (IC)

If the last residual $\hat{\varepsilon}_T = Y_T - \hat{Y}_T$ signals a break, add it to all future forecasts:

$$\tilde{Y}_{T+1|T}^{\text{ic}} = \underbrace{\hat{\alpha}Y_T + \hat{\mu}}_{\tilde{Y}_{T+1}} + \underbrace{(Y_T - \hat{\mu} - \hat{\alpha}Y_{T-1})}_{\hat{\varepsilon}_T} = Y_T + \hat{\alpha}(Y_T - Y_{T-1})$$

Differencing the estimated model

Difference the estimated model to remove level dependence:

$$\Delta \tilde{Y}_{T+1|T}^{\text{diff}} = \hat{\alpha} \Delta Y_T, \quad \tilde{Y}_{T+1|T}^{\text{diff}} = Y_T + \hat{\alpha} \Delta Y_T = \tilde{Y}_{T+1|T}^{\text{ic}}$$

The 1-step-ahead intercept-corrected forecast **equals** the 1-step forecast from the differenced model. Both are **level-robust**: they start from the current level Y_T rather than the modelled mean, so they adapt immediately to level shifts.

UK money demand: illustration of forecast failure and recovery

- Model estimated 1963–1985:3 with $R_{la,t}$ as cost of holding money
- Post-1985: own rate of money changes from zero $\Rightarrow$ location shift
- **Dynamic forecasts fail:** model predicts declining velocity but velocity actually falls faster
- **Static 1-step forecasts also fail:** $DW \approx 0$ in forecast period
- **Solution 1:** add level shift dummy from 1985:1 $\Rightarrow$ improved forecasts (equation 25 in Hendry & Ericsson, 1991)
- **Solution 2:** intercept-corrected forecasts using last 3 in-sample residuals $\Rightarrow$ best performance

Structural breaks at the forecast origin are the **norm**, not the exception, in economics and business. Always monitor residuals at the forecast origin and apply intercept correction when a break is detected.

Section 4

General-to-Specific Model Selection

Start from a **large** model with N candidate regressors that is well-specified (residuals are white noise, diagnostics pass). Then simplify.

$$Y_t = \sum_{j=1}^N \beta_j X_{j,t} + \varepsilon_t$$

A sub-model is **terminal** (or **pivotal**) if: (i) LR test vs. GUM is accepted; (ii) each retained variable is individually significant.

Why GETS rather than backward elimination?

- Standard backward elimination: sensitive to collinearity; can miss the true model
- GETS (Hoover & Perez, 1999; Hendry & Krolzig, 2001): **multipath search**
 - 1 Find 10 least significant regressors
 - 2 For each, eliminate and run backward elimination $\Rightarrow$ 10 paths
 - 3 Merge: retain variables significant across paths
- Implemented in **PcGets** (Hendry & Krolzig, 2005) and **Autometrics** (Doornik, 2009)

Autometrics + IIS: the full algorithm

The General Unrestricted Model includes:

- All candidate economic regressors
- Lagged dependent variables (up to $p_{\max}$)
- **Impulse indicator dummies** for every observation (IIS)
- Step indicator dummies (SIS) for every observation

Algorithm proceeds by:

- Splitting IIS dummies into blocks (e.g., first half, second half)
- Running GETS within each block
- Combining terminal models
- Final selection: retain variables significant at target level α

Gauge: fraction of irrelevant variables falsely retained. At target level $\alpha = 1\%$, gauge $\approx \alpha$ regardless of N .

Potency: fraction of relevant variables correctly retained. High potency requires signal-to-noise ratio β/σ to be large enough.

Autometrics achieves gauge $\approx \alpha$ even when $N > T$ (IIS case with T impulse dummies plus regressors).

UK velocity example (Hendry & Ericsson, 1991)

GUM: Δv_t on 11 lags of ΔC_t , 10 lags of Δv_t , ECM, plus $T = 94$ impulse dummies (IIS).

Autometrics (target size 1%) selects a single terminal model:

$$\widehat{\Delta v}_t = -0.097 \text{ecm}_{t-1} + 0.62 \Delta C_t - 0.28 \Delta v_{t-1} - 0.0621_{1971:1} + 0.0611_{1973:1} - 0.0511_{1974:4} + 0.0511_{1979:2}$$

- ECM: long-run adjustment to velocity equilibrium
- 4 impulse dummies: historically motivated interventions (decimalisation, fiscal expansion, GDP dip, end of Labour government)
- All diagnostics pass: no serial correlation, no ARCH, no heteroskedasticity, normality acceptable

IIS allowed detection of the 4 outlier observations without knowing their dates in advance — a fully data-driven approach with controlled false discovery rate.

Section 5

Penalised Methods and Machine Learning

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \left\{ \sum_{t=1}^T (Y_t - x'_t \beta)^2 + \lambda \sum_{j=1}^N |\beta_j| \right\}$$

The L_1 penalty drives some $\hat{\beta}_j$ **exactly to zero** $\Rightarrow$ automatic variable selection.

Ridge regression: uses L_2 penalty $\lambda \sum \beta_j^2 \Rightarrow$ shrinks but does not zero out.

LASSO for time series

- Choose λ by cross-validation (time-series CV: no data leakage)
- **Elastic net:** convex combination of L1 and L2 penalties
- **Adaptive LASSO:** weights each coefficient; consistent selection
- **Limitations:** assumes regressors on same scale;

Both select variables from a large set. Key differences:

- GETS: t-tests with controlled size; valid inference on retained variables
- LASSO: biased estimators; inference after selection is non-trivial

Classical vs. Machine Learning forecasting

M4 competition (Makridakis et al., 2018): 100,000 time series, 9 horizons, 61 methods.

Key findings:

- **Combination methods won**: averaging of exponential smoothing and ARIMA outperformed most individual models
- **Pure ML (neural networks) underperformed** relative to classical methods on standard series
- **Hybrid methods** (ML + ETS/ARIMA) performed best overall
- Simple methods (Theta, ETS) were remarkably competitive

Do not discard classical time series methods. They embody decades of understanding about series dynamics and are especially reliable for shorter horizons and smaller samples. Use ML to complement, not replace.

LASSO, Ridge, and Forecast Combination III

Forecast combination: practical implementation

Simple equal-weight combination:

$$\tilde{Y}_{T+h}^{\text{comb}} = \frac{1}{K} \sum_{k=1}^K \tilde{Y}_{T+h}^{(k)}$$

OLS combination weights (Bates & Granger, 1969):

$$\tilde{Y}_{T+h}^{\text{comb}} = \hat{w}_1 \tilde{Y}_{T+h}^{(1)} + \cdots + \hat{w}_K \tilde{Y}_{T+h}^{(K)}$$

where $\hat{w}$ minimises in-sample MSE. Constrain $\sum w_k = 1$, $w_k \geq 0$ for positivity.

If all component models share the same misspecification (e.g., all ignore a structural break), combination cannot help. Combination works best when models **differ structurally** — they make different errors in different periods.

Core references

- Diebold, F.X. & Mariano, R.S. (1995). Comparing predictive accuracy. *JBES*, 13, 253–263.
- Hendry, D.F. & Doornik, J.A. (2014). *Empirical Model Discovery*. MIT Press.
- Clements, M.P. & Hendry, D.F. (1999). *Forecasting Non-Stationary Economic Time Series*. MIT Press.
- Hyndman, R.J. & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd ed. OTexts. [free at otexts.com/fpp3]
- Hyndman, R.J. & Athanasopoulos, G. (2021). *Previsão: Princípios e Prática*. [otexts.com/fpp3pt]

Applied papers

- Makridakis, S., Spiliotis, E. & Assimakopoulos, V. (2018). Statistical and ML forecasting methods. *PLOS ONE*, 13, e0194889.
- Fildes, R. & Petropoulos, F. (2015). Simple vs. complex selection rules. *JBR*, 68, 1692–1701.
- Doornik, J. (2009). Autometrics. In Castle & Shephard (eds.), *Methodology and Practice of Econometrics*. OUP.
- Hendry, D.F. & Ericsson, N.R. (1991). Modeling narrow money demand. *European Economic Review*, 35, 833–881.

Software: R (gets: GETS/IIS; forecast: ARIMA, ETS, combination; opera: online forecast combination) or Python (statsforecast, darts, sktime). Applied exercises: DM test comparing ARIMA vs. ETS; run Autometrics IIS on a macroeconomic series; implement equal-weight and OLS combination.

Lecture 8 — Student Presentations

Seminar: Paper Presentations and Critical Discussion

Research question · Methodology · Results · Critical assessment · Class discussion